Test Plan Document

Blood Donation Management App

Project Code:

Internal Advisor:

Dr. Muhammad Ilyas

External Advisor:

Project Manager: **Shumaila Zafar**

Project Team: **Rameen Fatima**
Zobia Shaheen

Submission Date:

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Project Manager's Signature

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Author(s)	Shumaila Zafar, Rameen Fatima, Zobia Shaheen
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Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
ASP	Active Server Pages
TP	Test Plan
RTM	Requirements Traceability Matrix

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1. Introduction

The BDMS is designed to efficiently manage the entire process of blood donation from registration to blood distribution. This system is designed to reduce manual errors and for making timely availability of blood.

1.1 Purpose of Document

The purpose of this document is to test the BDMS, its functions and features. This document will be read by testing personnel, developers and managers.

1.2 Project Overview

The BDMS is designed to efficiently manage the entire process of blood donation from registration to blood distribution. This system is designed to reduce manual errors.

Software Usage:

- *This software will be used by donors, recipients, admins, hospitals and blood banks.*

Goals and Benefits:

- *The system aims to provide the efficient delivery of blood in normal and critical cases. It reduces manual errors, saves time and ensures timely availability of blood.*

2. Scope of Testing

The BDMS is tested only for donor-recipient management, blood group matching, data handling and admin functionalities while payments and emergency handling are out of scope.

What will be tested in BDMS:

- *Donor registration and login*
- *Recipient registration and login*
- *Admin registration and login*
- *Login authentication*
- *Adding new donors*
- *Updating donors' data*
- *Verification of users*
- *Searching for donors by*
  1. *Blood group*
  2. *City, Area*
  3. *Availability status*

What will be tested in BDMS:

- *Medical verification (Medical eligibility of donors)*
- *live connection with hospitals and blood banks*
- *Real time blood stock update*
- *Live tracking of donation*

3. Test Plan Strategy

The following test plan is therefore primarily designed to confirm that BDMS meets the requirements defined in the SRS. Manual testing has been used in this project. At each stage testing is done such that any error detected might be corrected before the next stage is approached.

3.1 Unit Testing

Definition:

Unit testing checks individual modules of the system separately. For example, registration, login and search functions.

Participants: *Project team members*

Methodology:

It follows the following approach:

- *Each module is tested separately.*
- *Both valid and invalid inputs are used.*
- *Errors are corrected before integrating modules.*

3.2 Integration Testing

Definition:

The goal of integration testing is to ensure that various modules interact properly with each other.

Participants:

Project team

Methodology:

It follows the following approach:

- *Combine related modules*
- *Check data flow between donor, recipient and admin modules*

3.3 System Testing

Definition:

System testing checks the complete system to ensure that it meets the requirements mentioned in SRS.

Participants:

Testing team

Methodology:

It follows the following approach:

- *Perform end to end test cases*
- *Restate all functional and non-functional requirements*
- *Check system behavior for normal and wrong input*

4. Test Environment

Hardware:

- *Personal computer (Desktop or Laptop)*
- *Smart Phone (Android or IOS)*

Software:

- *Window 10*
- *C++ Compiler*

Network:

- *Stable network required*

5. Schedule

Testing Activities	Begin	End
Designing Test Cases	01-12-2025	03-12-2025
Executing Test Cases	04-12-2025	05-12-2025
Unit Testing	06-12-2025	07-12-2025
Integration Testing	08-12-2025	09-12-2025
System Testing	10-12-2025	10-12-2025

6. Control Activities

- *Regular review meetings with the project advisor*
- *Test case review before execution and tracking of identical defects*

7. Functions to be Tested

This section describes the key abilities that are evaluated in BDMS to ensure they meet the SRS specifications and that they are well implemented according to the design plans.

7.1. User registration

- *Donor registration.*
- *Receiver registration.*
- *Validation of mandatory fields: Name, ID, Blood group*

7.2. User login and Authentication

- *Login with valid credentials.*
- *Handling invalid login attempts.*
- *Role-based access: Donor, Recipients, Admin.*

7.3. Donor Management

- *Inserting new records of donors.*
- *Updating status of donor availability.*
- *Access to donor profile information.*
- *Blood group search amongst potential blood donors.*

7.4. Receiver Management

- *Creating blood requests.*
- *Status of blood request.*
- *Cancelation of blood request.*

7.5. Blood Request Management

- *Create Blood request.*
- *Admin view all the requests.*
- *Requests acceptance or cancellation by admin.*
- *Status Updates: pending, accepted or cancelled.*

7.6. Blood Stock Management

- *Showing the available blood stock.*
- *Updating blood stock levels.*
- *Check stocks by blood group.*

7.7. Appointments Management

- *Scheduling donor appointments.*
- *Displaying appointments information.*

7.8. Administrative Functions

- *Verification of users.*
- *Viewing all users.*
- *Generating basic system report.*

8. Functions not to be Tested

Testing out of scope includes the following functions, which are either not included in the present system requirements or scheduled for future versions.

8.1. Medical verification of Donor

- *No medical eligibility check.*
- *Reason: Require medical systems and expert validation.*

8.2. Real-time Integration of Hospitals and Blood Banks

- *No live interactions with any hospitals and blood banks.*
- *Reason: External system is not integrated into this system.*

8.3. Online Payments or Financial Transactions

- *No functionality for payment or billing exists.*
- *Reason: Not mention within the SRS and design scope.*

8.4. Real time Blood Stock Updates

- *The stocks are updated manually.*
- *Reason: It is out of stock with the present system to allow live tracking.*

8.5. Emergency Handling and Alarms

- *Emergency notifications are not included.*
- *Reason: Planned as future enhancements.*

9. Test Case Design and Description

This section describes the detailed test cases designed to verify that the Blood Donation Management System (BDMS) functions according to the requirements specified in the Software Requirements Specification (SRS).

Each test case focuses on validating a specific functional module of the system, including user registration, authentication, donor management, blood request handling, blood stock management, and administrative operations.

The test cases are designed using a black box testing approach, where the internal code structure is not considered and testing is performed based on expected inputs and outputs. Both positive and negative scenarios are included to ensure that the system behaves correctly under valid conditions and handles invalid inputs gracefully. Each test case includes:

- *Test Case ID*
- *Objective*
- *Pre-requisites*
- *Test steps*
- *Expected results*
- *Execution status*

These test cases ensure complete functional coverage of the system and help identify defects at early stages, improving overall system quality and reliability.

12.1. Test Case 1
Donor Registration

Test Case ID:	*TC-01*	**QA Test Engineer**	*Project Team*
Test case Version:	*1.0*	**Reviewed By**	*Testing Team lead*
Test Date:	*03-12-2025*	**Use Case Reference(s)**	*UC-01 (Donor Registration)*
Revision History	*New*		
Objective	*To verify that a donor can successfully register by providing valid and complete information.*		
Product/Ver/Module	*Blood Donation Management System / Version 1.0 / Donor Registration Module*		
Environment:	*Hardware: PC or Laptop* *Software: Windows 10, C++ Compiler*		
Assumptions:	*The donor is not already registered in the system.*		
Pre-Requisite:	*Application is running and database connection is active.*		

Step No.	**Execution description**	**Procedure result**
1. 2. 3.	*Open donor registration form* *Enter valid donor details* *Click register button*	*Registration form displayed* *Data accepted* *Donor registered successfully*
	Passed	
	☐ Passed ☐ Failed ☐ Not Executed	

12.2. Test Case 2

Test Case ID:	*TC-02*	**QA Test Engineer**	*Project Team*
Test case Version:	*1.0*	**Reviewed By**	*Testing Team lead*
Test Date:	*03-12-2025*	**Use Case Reference(s)**	*UC-01 (Donor Registration)*
Revision History	*New*		
Objective	*To validate that system displays an error message when mandatory fields are missing.*		
Product/Ver/Module	*Blood Donation Management System / Version 1.0 / Donor Registration*		
Environment:	*Hardware: PC or Laptop* *Software: Windows 10, C++ Compiler*		
Assumptions:	*User attempts to submit incomplete data.*		
Pre-Requisite:	*Registration page is opened.*		

Step No.	**Execution description**	**Procedure result**
1. 2.	*Leave blood group empty* *Click register button*	*Field remains empty* *Error message displayed*

Passed	
	☐ Passed ☐ Failed ☐ Not Executed

12.3. Test Case 3

Test Case ID:	*TC-03*	**QA Test Engineer**	*Project Team*
Test case Version:	*1.0*	**Reviewed By**	*Testing Team lead*
Test Date:	*04-12-2025*	**Use Case Reference(s)**	*UC-02 (User login)*
Revision History	*New*		
Objective	*To verify that a registered user can log in with valid credentials.*		
Product/Ver/Module	*Blood Donation Management System / Version 1.0 / Authentication Module*		
Environment:	*Hardware: PC or Laptop* *Software: Windows 10, C++ Compiler*		
Assumptions:	*User is already registered.*		
Pre-Requisite:	*Valid username and password exist.*		

Step No.	**Execution description**	**Procedure result**
1. 2. 3.	*Enter valid username* *Enter valid password* *Click Login*	*Username accepted* *Password accepted* *User logged in successfully*

Passed	
	☐ Passed ☐ Failed ☐ Not Executed

12.4. Test Case 4

Test Case ID:	*TC-04*	**QA Test Engineer**	*Project Team*
Test case Version:	*1.0*	**Reviewed By**	*Testing Team lead*
Test Date:	*04-12-2025*	**Use Case Reference(s)**	*UC-02 (User login)*

Revision History	*New*	
Objective	*To verify system behavior when invalid login credentials are entered.*	
Product/Ver/Module	*Blood Donation Management System / Version 1.0 / Authentication Module*	
Environment:	*Hardware: PC or Laptop* *Software: Windows 10, C++ Compiler*	
Assumptions:	*User enters incorrect password.*	
Pre-Requisite:	*Login page is displayed.*	
Step No.	**Execution description**	**Procedure result**
1. 2. 3.	*Enter valid username* *Enter wrong password* *Click Login*	*Username accepted* *Password rejected* *Error message shown*
Passed		
	☐Passed ☐Failed ☐Not Executed	

12.5. Test Case 5

Test Case ID:	*TC-05*	**QA Test Engineer**	*Project Team*
Test case Version:	*1.0*	**Reviewed By**	*Testing Team lead*
Test Date:	*05-12-2025*	**Use Case Reference(s)**	*UC-03 (Search Donor)*
Revision History	*New*		
Objective	*To verify that donors can be searched based on blood group.*		
Product/Ver/Module	*Blood Donation Management System / Version 1.0 /Donor Management*		
Environment:	*Hardware: PC or Laptop* *Software: Windows 10, C++ Compiler*		
Assumptions:	*Donor records exist in the database.*		
Pre-Requisite:	*User is logged in.*		
Step No.	**Execution description**		**Procedure result**
1. 2.	*Select blood group* *Click search*		*Selection accepted* *Matching donors displayed*
Passed			
	☐Passed ☐Failed ☐Not Executed		

10. Traceability Matrix

The Requirements Traceability Matrix (RTM) establishes a clear relationship between system requirements, use cases, and test cases. The purpose of the traceability matrix is to ensure that every functional requirement defined in the SRS is properly tested and that no requirement is left unverified.
This matrix supports both:

- *Forward traceability (from requirements to test cases), ensuring all requirements are tested.*
- *Backward traceability (from test cases to requirements), ensuring that each test case maps to a valid requirement.*

The RTM helps in:

- *Tracking testing progress*
- *Identifying missing test coverage*
- *Assessing the impact of requirement changes*
- *Ensuring quality assurance and compliance with project objectives*

A requirement is marked as Complete once all associated test cases have been successfully executed and passed.

Sr. no.	Requirement ID	Use Case ID	GUI ID	Test case ID	Status	Date Completed
1	FR-01 (Donor Registration)	UC-01	GUI-01	TC-01	Complete	03-12-2025
2	FR-01 (Donor Registration Validation)	UC-01	GUI-01	TC-02	Complete	03-12-2025
3	FR-02 (User Login)	UC-02	GUI-02	TC-03	Complete	04-12-2025
4	FR-01 (Invalid Login Handling)	UC-02	GUI-02	TC-04	Complete	04-12-2025
5	FR-01 (Search Donor by Blood Group)	UC-03	GUI-03	TC-05	Complete	05-12-2025

11. Major Deliverables

The following deliverables will be produced at the end of the testing phase of the Blood Donation Management System (BDMS):

- ***Test plan***
  A complete test plan defining testing scope, objectives, testing strategy, resources, schedule, risks and exit criteria for BDMS.
- ***Test cases***
  Detailed test cases covering all functional modules of BDMS including donor management, blood inventory, requests, authentication and reporting.
- ***Test summary report***
  A final report highlighting testing outcomes, quality assessment of BDMS, known issues and recommendations for release.

12. Risks and Assumptions

Assumptions:

The following assumptions were considered while planning and executing testing for BDMS:

- *Unit testing has been completed before system testing begins.*
- *The requirements document is stable, complete and approved.*

- *Testing personnel have sufficient knowledge of BDMS and basic testing components.*
- *Required tools are available during the testing phase.*
- *Test environment was suitable for testing.*

Risks:

- *Requirements' creeping.*
- *Ambiguious requirements.*
- *Limited availability of resources for testing (time, money, members).*
- *Delays in bug fixing by the development team.*

13. Exit Criteria

The testing document will be considered complete when the following conditions are met:

- *All planned test cases have been executed.*
- *All errors are fixed.*
- *All core functionalities (donor registration, blood requests, inventory management, authentication and reporting) work as per requirements.*
- *Test summary report is prepared and approved by stakeholders.*
- *The system is stable and ready for deployment.*

14. References

- *Course Instructor Guidelines*
- *Blood Donation Management System (BDMS). (2025) Software Requirements Specifications (SRS)*
- *Blood Donation Management System (BDMS). (2025) Systems Design Document (SDD)*
- *Roger S. Pressman, Software Engineering*